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Creating a universal cell segmentation algorithm

Cell segmentation currently involves the use of various bespoke algorithms designed for specific cell types, tissues, staining methods and microscopy technologies. We present a universal algorithm that can segment all kinds of microscopy images and cell types across diverse imaging protocols.

The problem

Microscopy is a fundamental pillar of biological research, enabling scientists to understand structures and functions at the cellular level. With the continuing emergence of new imaging technologies, there is a growing demand for fully automated cell segmentation with algorithms that are capable of unsupervised identification of individual cells in microscopy images — a key step for quantitative cell analysis and tracking and for evaluating cellular reactions in cancer research and drug discovery.

However, cell segmentation is complicated by the wide variety of cell origins and morphologies, staining methods and types of microscope. Existing algorithms are typically tailored for specific cell types and imaging techniques and often require hyper-parameter adjustments to optimize performance, presenting biologists with challenges in selecting the right tool for their images. This requirement to identify appropriate algorithms highlights the urgent demand for a universal, user-friendly cell segmentation algorithm that is capable of handling various microscopy images without additional manual intervention[1,2].

The solution

To tackle this challenge, we initiated a global competition at the Neural Information Processing Systems (NeurIPS) machine learning conference to harness the resourcefulness of the machine learning community to expedite algorithm development, a strategy that has previously proved effective[3,4]. We curated a large-scale dataset with over 3,000 microscopy images from more than 50 different biological experiments, collected from more than 20 biology laboratories (Fig. 1). We created a hidden testing set with previously unseen images to benchmark the generalization ability of the algorithms. The competition attracted hundreds of participants from all over the world, and we invited the top 30 teams, as ranked by the performance of their algorithm on the public tuning set, to make the testing submission by packaging their algorithms as standardized, standalone executable software. We executed the submitted software packages sequentially on the same workstation, evaluating both segmentation accuracy (F1 scores, an average of precision and recall that ranges between 0 and 1, the best score) and efficiency (computational running time) on the large-scale challenge dataset, to establish a final ranking.

Our analysis shows that algorithms using the transformer-based architecture markedly outperformed traditional convolutional neural network-based algorithms. The winning algorithm has not only superior generalization ability but also remarkable efficiency, and it can be directly applied to a wide range of microscopy images without manual hyper-parameter adjustments. Furthermore, the top-performing teams have incorporated their algorithms into user-friendly tools, such as napari, narrowing the gap between development of these advanced algorithms and their deployment in daily biological practice. This endeavor not only paves the way for the wider adoption of advanced computational methods in routine biological research but also signifies a pivotal advance in creating foundation models for bioimage analysis.

The implications

These new cell segmentation algorithms have many potential applications across a spectrum of tasks. For example, cell segmentation is a key step in characterizing cellular phenotypes in tumor tissue, thereby uncovering complex cellular structures. The algorithms could also have an important role in disease progression studies, particularly in characterizing cancer microenvironments. In addition, we aim to establish this competition as an evolving benchmarking platform that incorporates a greater diversity of data, paving the way for continuous advancement in this vital field.

While the challenge dataset we used is diverse, it mainly contains 2D microscopy images. However, the rising prevalence of 3D microscopy images presents new challenges for segmentation algorithms, such as the high dimensionality of the datasets and their inherently low signal-to-noise ratio.

Looking to the future, our goal is to foster the development of foundation models for general biological images by expanding the benchmark with more imaging modalities and cellular structures, especially 3D images (for example, electron microscopy and cryo-electron tomography data)[5]. We believe that this can be accomplished with deep interdisciplinary collaborations between biologists and computer scientists worldwide.

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EXPERT OPINION

"The cell segmentation challenge on NeurIPS extends generalist cell segmentation algorithms to deal with multimodal images, including brightfield, phase contrast and differential interference contrast images, in addition to fluorescence images. The authors also analyze the performance of individual algorithms by using two metrics (accuracy and efficiency) to select the optimal algorithm. It is interesting that the best-performing algorithm achieved a great advancement, providing a proof of concept of obtaining novel algorithms by harnessing the collective expertise of biological imaging and machine learning experts." **Yi Wang, Zhejiang University, Hangzhou, China.**

FIGURE

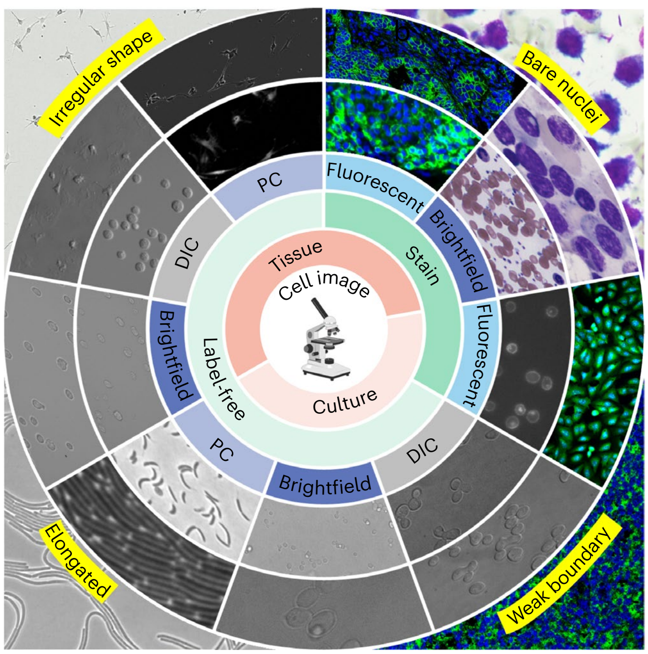

Fig. 1 | Diversity of microscopy images in the challenge dataset for assessing cell segmentation algorithms. The schematic depicts the diversity of the microscopy images dataset used to test the algorithms, including various sample origins, cell types and features, staining methods, and microscopy technologies. PC, phase contrast; DIC, differential interference contrast. © 2024, Ma, J. et al.

BEHIND THE PAPER

Cell segmentation is usually the first step in microscopy image-based biological research. On entering this field, I encountered the challenge of selecting appropriate algorithms from numerous segmentation tools, many of which require manual adjustment of hyper-parameters to achieve optimal performance. This experience led me to wonder whether it was possible to build a versatile cell segmentation algorithm that can handle all kinds of microscopy images. Developing a generalist model necessitates a large, diverse dataset that no single laboratory can produce in isolation. This realization sparked the idea of fostering collaboration among researchers to collectively address this challenge, including the support of many biological researchers. To attract researchers in artificial intelligence, we organized this competition within a machine learning conference, producing a winning solution with strong performance and efficiency. This journey exemplifies the profound impact of multidisciplinary collaboration in advancing science. **J.M.**

REFERENCES

1. Moen, E. et al. Deep learning for cellular image analysis. *Nat. Methods* **16**, 1233–1246 (2019).
   A review article that highlights the great potential of deep learning for cell segmentation in microscopy images.
2. Stringer, C. et al. Cellpose: a generalist algorithm for cellular segmentation. *Nat. Methods* **18**, 100–106 (2021).
   This paper presents a generalist model for cell segmentation.
3. Caicedo, J. C. et al. Nucleus segmentation across imaging experiments: the 2018 Data Science Bowl. *Nat. Methods* **16**, 1247–1253 (2019).
   This paper demonstrates that international competition is an effective way to solve challenging biological tasks.
4. Maška, M. et al. The Cell Tracking Challenge: 10 years of objective benchmarking. *Nat. Methods* **20**, 1010–1020 (2023).
   This paper summarizes the challenges of cell segmentation and tracking.
5. Ma, J. & Wang, B. Towards foundation models of biological image segmentation. *Nat. Methods* **20**, 953–955 (2023).
   This commentary article describes a blueprint for developing foundation models for bioimage segmentation.

FROM THE EDITOR

"Segmentation is a critical task in contemporary image analysis that can be facilitated by numerous computational tools. The performance of these tools can be challenging to assess, especially in the absence of common benchmarking. This benchmarking paper caught my eye not only for the scope and quality of the comparisons, but also because it sets the stage for transformer-based deep learning models to become the new gold standard." **Rita Strack, Senior Editor, *Nature Methods*.**